Duty at home

DM1 · One sum and one threshold · 45 min, after S11.

What's next? (u_n) defined by $u_0 = 2$ and, for everything $n \in \mathbb{N}$, $u_{n+1} = \frac{1}{2}u_n + 3$.

1. Demonstrate by induction that $2 \leq u_n < 6$ for all $n \in \mathbb{N}$. Deducing the direction of variation of (u_n).

2. Set $v_n = u_n - 6$ for all $n \in \mathbb{N}$. Show that (v_n) is geometric, then express u_n in function of n.

3. For all $n \in \mathbb{N}$, we note $S_n = \sum_{k=0}^{n} u_k$. Express S_n according to the n.

4. Using the explicit formula and powers of 2, determine the smallest integer n such that $u_n > 5.99$. Check that the integer previous one is not suitable.

5. Express $6 - \frac{S_n}{n+1}$ and justify that this difference is strictly positive. Interpret the result in Whereas the average $n + 1$ first terms; no limit calculation is required.